

About This Document

The technical round covers SQL, databases, warehousing, ETL/pipelines, big data and the cloud stack (Snowflake, Azure) BI3 uses.

Focus areas (research-backed): SQL fundamentals (joins, GROUP BY/HAVING, second-highest-salary) and data-modeling basics are central to the technical round.

SQL and Databases

8 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Which clause filters rows AFTER aggregation?

(A) WHERE

(B) HAVING ✓

(C) GROUP BY

(D) ORDER BY

Answer: (B) HAVING

Authenticity: Frequently Repeated

Q2. Which JOIN keeps all left rows plus matches (NULLs otherwise)?

(A) INNER JOIN

(B) LEFT JOIN ✓

(C) CROSS JOIN

(D) SELF JOIN

Answer: (B) LEFT JOIN

Authenticity: Community Reported

Q3. A PRIMARY KEY guarantees values are:

(A) Sorted

(B) Unique and not null ✓

(C) Indexed only

(D) Foreign

Answer: (B) Unique and not null

Authenticity: Community Reported

Q4. Which normal form removes partial dependency on part of a composite key?

(A) 1NF

(B) 2NF ✓

(C) 3NF

(D) BCNF

Answer: (B) 2NF

Authenticity: Community Reported

Q5. Which quickly removes all rows and cannot be easily rolled back?

(A) DELETE

(B) TRUNCATE ✓

(C) DROP

(D) REMOVE

Answer: (B) TRUNCATE

Authenticity: Community Reported

Q6. Which is an aggregate function?

(A) UPPER

(B) LENGTH

(C) AVG ✓

(D) SUBSTR

Answer: (C) AVG

Authenticity: Community Reported

Q7. Which finds the second highest salary correctly?

(A) SELECT MAX(salary) FROM emp

(B) SELECT MAX(salary) FROM emp WHERE salary < (SELECT MAX(salary) FROM emp) ✓

(C) SELECT salary FROM emp LIMIT 2

(D) SELECT MIN(salary) FROM emp

Answer: (B) SELECT MAX(salary) FROM emp WHERE salary < (SELECT MAX(salary) FROM emp)

Authenticity: Verified Previous-Year

Q8. A FOREIGN KEY is used to:

(A) Sort rows

(B) Reference another table's primary key ✓

(C) Encrypt data

(D) Index a column

Answer: (B) Reference another table's primary key

Authenticity: Community Reported

Data Warehousing & Modeling

6 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. The central table holding measures is the:

(A) Dimension table

(B) Fact table ✓

(C) Lookup table

(D) Bridge table

Answer: (B) Fact table

Authenticity: Community Reported

Q2. A central fact table with denormalized dimensions is a:

(A) Snowflake schema

(B) Star schema ✓

(C) Galaxy schema

(D) Network schema

Answer: (B) Star schema

Authenticity: Community Reported

Q3. A snowflake schema has dimension tables that are:

(A) Denormalized

(B) Normalized into sub-dimensions ✓

(C) Removed

(D) Combined

Answer: (B) Normalized into sub-dimensions

Authenticity: Community Reported

Q4. OLAP systems are optimized for:

(A) High-volume writes

(B) Analytical queries and aggregation ✓

(C) Row locking

(D) Payments

Answer: (B) Analytical queries and aggregation

Authenticity: Community Reported

Q5. SCD Type 2 handles a change by:

(A) Overwriting old value

(B) Adding a new versioned row ✓

(C) Deleting the record

(D) Ignoring it

Answer: (B) Adding a new versioned row

Authenticity: Community Reported

Q6. A data warehouse stores data that is:

(A) Real-time only

(B) Subject-oriented, integrated, historical ✓

(C) Unstructured only

(D) Temporary

Answer: (B) Subject-oriented, integrated, historical

Authenticity: Community Reported

ETL, Pipelines & Big Data

6 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. ETL stands for:

(A) Extract, Transform, Load ✓

(B) Export, Transfer, Link

(C) Extract, Transfer, Log

(D) Evaluate, Transform, Load

Answer: (A) Extract, Transform, Load

Authenticity: Frequently Repeated

Q2. In ELT, transformation happens:

(A) Before extraction

(B) After loading into the target ✓

(C) Never

(D) During extraction only

Answer: (B) After loading into the target

Authenticity: Community Reported

Q3. Which model handles data in real time as it arrives?

(A) Batch

(B) Stream processing ✓

(C) Manual

(D) Nightly

Answer: (B) Stream processing

Authenticity: Community Reported

Q4. Apache Spark processes data mainly via:

(A) Disk-only MapReduce

(B) In-memory distributed computing ✓

(C) Single-threaded loops

(D) Stored procedures

Answer: (B) In-memory distributed computing

Authenticity: Community Reported

Q5. Idempotency means re-running a step:

(A) Doubles data

(B) Produces the same result without duplicates ✓

(C) Fails

(D) Deletes source

Answer: (B) Produces the same result without duplicates

Authenticity: Community Reported

Q6. Partitioning a large table helps by:

(A) Increasing cost

(B) Improving query performance via pruning ✓

(C) Removing indexes

(D) Preventing joins

Answer: (B) Improving query performance via pruning

Authenticity: Community Reported

Cloud & Snowflake

5 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Snowflake separates storage and compute so they:

(A) Scale together

(B) Scale independently ✓

(C) Are disabled

(D) Are limited

Answer: (B) Scale independently

Authenticity: Community Reported

Q2. The Snowflake compute resource that runs queries is a:

(A) Stage

(B) Virtual warehouse ✓

(C) Micro-partition

(D) Role

Answer: (B) Virtual warehouse

Authenticity: Community Reported

Q3. A data lake stores:

(A) Only relational data

(B) Raw data in many formats ✓

(C) Only reports

(D) Only images

Answer: (B) Raw data in many formats

Authenticity: Community Reported

Q4. A columnar file format common in analytics:

(A) CSV

(B) Parquet ✓

(C) TXT

(D) XML

Answer: (B) Parquet

Authenticity: Community Reported

Q5. Amazon S3 / Azure Blob are best described as:

(A) Relational DBs

(B) Scalable object/file storage ✓

(C) Message queues

(D) Compute clusters

Answer: (B) Scalable object/file storage

Authenticity: Community Reported

Python for Data Engineering

5 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Library most used for tabular data manipulation:

(A) NumPy

(B) pandas ✓

(C) matplotlib

(D) requests

Answer: (B) pandas

Authenticity: Community Reported

Q2. pandas method to drop rows with NaN:

(A) fillna()

(B) dropna() ✓

(C) isnull()

(D) merge()

Answer: (B) dropna()

Authenticity: Community Reported

Q3. pandas operation like a SQL join:

(A) concat()

(B) merge() ✓

(C) append()

(D) pivot()

Answer: (B) merge()

Authenticity: Community Reported

Q4. Python structure with unique unordered elements:

(A) list

(B) tuple

(C) set ✓

(D) dict

Answer: (C) set

Authenticity: Community Reported

Q5. groupby() is usually followed by:

(A) A sort only

(B) An aggregation like sum()/mean() ✓

(C) Deletion

(D) Nothing

Answer: (B) An aggregation like sum()/mean()

Authenticity: Community Reported

BI3 Technologies

Data Engineering Concepts (MCQ)

Applied Role(s):

[Data Engineer - Trainee](#)

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.